Ark

Money as Shared Infrastructure

Abstract

Ark is a monetary system built as shared infrastructure: fiat-tracking currencies created and redeemed through a native asset, NOAH, within an institution its participants can help own and govern. Building on Terra's design and the lessons of its collapse, Ark treats sustaining NOAH's value and liquidity as fundamental to the monetary system.

Its approach rests on three complementary principles: a founding philosophy grounded in the continuing search for truth, stability across changing conditions, and shared participation and ownership. These principles seek to sustain commitment to the network, confidence in its institutions, and economic use beyond its early supporters.

Expansion replenishes a Redemption Buffer, Strategic Reserve, and Insurance fund with distinct purposes. Defined rules govern monetary distress, while fees and a separate subsidy pool fund network operation. Ownership and governance are intended to broaden over time, allowing participants to shape Ark's development without requiring agreement with its founding philosophy.

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1 Introduction

Money allows people to cooperate without first agreeing on how the world should be organised. They can exchange goods, pay for work, and make commitments to one another while holding different beliefs and pursuing different ends. Yet the money that enables this cooperation also connects them through a shared system. The question is therefore not only how a currency is issued, but what kind of institution people come to depend on through its use.

Ark is an attempt to build that institution as shared infrastructure. Its purpose is to provide currencies that support ordinary economic activity within a network that its participants can help own and govern. Participation should be worthwhile because of what the system enables, without requiring agreement with its founders. Its long-term direction should be shaped by those who come to use and maintain it, rather than remain permanently under founding control.

Ark builds on Terra's original proposition of algorithmic stability: currencies that track familiar fiat units can expand and contract with demand through conversion with a native asset. Rather than requiring equivalent fiat reserves for every unit issued, the system relies on market incentives to maintain stability. Ark retains this approach, using NOAH as its native asset, because it allows monetary supply to respond directly to demand [1].

But flexibility in supply is not the same as stability in value. Terra's experience exposed a central limitation of elastic conversion: a protocol can issue its native asset to meet redemptions, but it cannot ensure that the market will absorb it at a value sufficient to preserve the currency's purchasing power. For Ark, sustaining NOAH's value and liquidity is therefore fundamental to the viability of the monetary system. The central challenge is to establish lasting reasons to acquire and hold NOAH, while managing the pressures that monetary expansion and contraction place upon it.

Ark approaches this challenge through three complementary principles.

The first is its founding philosophy. Ark emerged from the ideas developed in *The Revolution Begins Within*, a manifesto grounded in the continuing search for truth, openness to different perspectives, and the responsibility to remain willing to revise our understanding. These principles offer a purpose for building Ark that extends beyond financial returns. They are not beliefs that participants must accept, but foundations for an institution intended to remain open to people who understand the world differently.

The second is stability across changing conditions. Monetary incentives, available resources, institutional decisions, and public confidence affect one another. A system that functions during expansion may become vulnerable when those relationships reverse. Growth should therefore strengthen its capacity to withstand contraction, rather than become an assumption on which continued operation depends. This requires both resources prepared for future stress and procedures for situations in which ordinary mechanisms become insufficient.

The third is shared participation and ownership. Ark seeks to become a neutral monetary system whose usefulness does not depend on the authority or beliefs of its founders. By distributing ownership among members and contributors, and progressively opening governance to their participation, Ark aims to establish a broad community with a shared interest in its continued success. Its long-term value should emerge from what people can do with the system and the relationships they establish through its use.

These foundations address different parts of the same challenge. The founding philosophy offers a reason to commit resources and effort to Ark, alongside a standard for how the institution should behave. Its stability arrangements seek to preserve confidence through changing conditions by preparing resources, governing their use, and establishing how monetary distress is handled. Shared participation and ownership broaden the opportunity to acquire a stake in the network and shape its development. Together, these principles are intended to sustain the commitment, confidence, and economic activity on which NOAH's value—and the monetary system it supports—depend.

2 Lessons from Terra

2.1 The original monetary model

Terra's 2019 whitepaper proposed currencies that would track regional fiat units through conversion with Luna. Luna would absorb changes in currency demand: redemptions during contraction would increase its supply and dilute existing holders. The design recognised this exposure and proposed adjustable fees and seigniorage allocation to support validator compensation, alongside funding for applications intended to encourage adoption [1].

Adoption was central to the argument. Growing currency demand could reduce Luna's supply and support its market value, while expansion proceeds could finance further development. But neither scarcity nor currency circulation guaranteed lasting demand for Luna itself. The protocol could issue Luna to meet redemptions; it could not ensure that buyers would absorb that issuance at a sufficient price.

2.2 Policy during expansion

During Terra's expansion, policy increasingly favoured burning Luna and reducing costs that might discourage adoption. In March 2021, a proposal argued that community and oracle reward pools were overfunded and recommended burning accumulated community funds, directing future seigniorage towards burning, and extending distribution of existing oracle rewards. The proposal aimed to simplify the relationship between currency growth and Luna scarcity by replacing adjustable seigniorage allocation with a burn policy [2].

The Columbus-5 upgrade implemented the burn policy and redirected swap fees to oracle rewards [3, 5]. In January 2022, another proposal sought to eliminate the transfer-based stability fee, citing implementation difficulties and its limited revenue contribution [4]. Terra's historical documentation records that its original Treasury policy levers were no longer economically active: seigniorage was burned and the stability fee was fixed at zero [5].

Burning Luna and reducing transaction costs were not inherently harmful, and these decisions did not create the conversion model's underlying vulnerability. The relevant lesson is narrower: policies that made the native asset more attractive during expansion did not necessarily preserve resources or independent funding through contraction. Ark's retention rules respond to that distinction, rather than treating every growth-oriented decision as a mistake.

2.3 Conversion capacity and market demand

Increasing redemption capacity did not necessarily increase the market's willingness to buy the Luna issued in exchange. In January 2022, Jump Trading proposed a virtual pool of 50 million SDR and a recovery period of 36 blocks, estimating daily mint-and-burn capacity of approximately $293 million. The proposal sought to accommodate growing activity while considering oracle-manipulation risk [6].

During the May 2022 depeg, a further proposal sought a 100 million SDR pool and an 18-block recovery period, raising estimated daily capacity to approximately $1.2 billion. It recognised the conflict between protecting UST and protecting Luna. These were estimates of protocol conversion capacity, not guarantees of market liquidity [7].

When sufficient buyers existed, redemptions could support arbitrage towards UST's intended value. When confidence deteriorated, additional Luna could intensify selling pressure, depress its price, and require more issuance for subsequent redemptions. Faster conversion could therefore accelerate the feedback loop it was intended to resolve.

Terra's response was not limited to elastic conversion. A forensic analysis by JS Held, commissioned through counsel and paid for by Terraform Labs, documents the use of LFG's Bitcoin and stablecoin reserves in the May 2022 peg defence. Reserve intervention occurred, but did not preserve the peg [8]. The comparison with Ark is therefore about how resources are accumulated, recognised, committed, and governed—not a claim that Terra had no reserves or attempted no intervention.

2.4 The fragility of subsidised adoption

Growing currency circulation can reflect lasting economic use or a temporary response to incentives. Anchor became a major source of demand for UST by offering deposit returns supported in part through subsidies. Liu, Makarov, and Schoar document repeated replenishment of Anchor's yield reserve, including funding obtained by converting Luna into UST. Growing deposits could therefore coexist with a recurring funding deficit [9].

The problem was recognised before the collapse. In March 2022, Anchor considered a deposit rate that would adjust with its yield reserve. Lower returns could reduce the funding burden, but could also weaken the reason depositors remained [10].

The distinction is between attracting demand and sustaining it. Ark offers different reasons to participate, including shared ownership and bounded compensation for eligible losses. These benefits can encourage adoption, but their contribution to lasting demand depends on whether participants continue to find value in Ark's currencies and whether the institution can sustain the resources required to provide them.

2.5 Resolution after collapse

Terra's eventual revival plan proposed a new blockchain without the algorithmic stablecoin and a distribution of new tokens among affected groups. The plan's terms were negotiated after the collapse through a new governance proposal [11].

When a monetary mechanism fails, decisions remain about outstanding balances, available resources, and the allocation of losses. A predefined process cannot recover value merely by changing the accounting, but it can make the available actions and their consequences

visible before a crisis. Ark separates suspension, fixed-rate settlement, write-off, and retirement for that reason.

2.6 Design implications for Ark

Terra's experience makes the relationship between expansion, redemption, and native-asset demand central to Ark's design. Increasing conversion capacity cannot substitute for market liquidity, and attracting currency deposits does not by itself establish lasting reasons to use the system. Ark therefore treats capital policy, operating finance, and the sources of participation as related parts of monetary stability.

Expansion retains NOAH under distinct commitments: the Buffer funds a proportional share of redemptions, the Reserve preserves resources for discretionary intervention, and Insurance finances approved compensation. Fees and a separate subsidy pool support network operation. Currency states define the available responses when ordinary conversion becomes unsustainable, including how settlement and unpaid obligations are handled.

The following sections explain these arrangements alongside the institution's founding purpose and distribution of authority. Shared purpose can motivate people to begin building and using Ark; continued participation must also be supported by the value its currencies and institutions provide.

3 Philosophy and monetary confidence

Money depends in part on collective confidence. People accept a currency because they expect others to accept it, and that expectation reflects confidence not only in its economic usefulness, but also in the institutions and arrangements that sustain its value. For Ark, confidence in NOAH is therefore inseparable from confidence in the institution behind it.

Ark seeks to build that confidence around its founding philosophy. People who recognise the value of its principles may find in Ark an institution whose purpose they believe in and whose future they wish to support. Their conviction can become a source of economic commitment, while the institution's continued adherence to those principles provides a basis on which that confidence can be sustained. Ark's philosophy is therefore more than a statement of purpose: it is intended to become part of the foundation of its monetary confidence.

3.1 Founding philosophy

Ark emerged from the ideas developed in *The Revolution Begins Within*. At the heart of the manifesto is the belief that our understanding of the world is never complete, and that the search for truth requires us to remain open to questioning, correction, and perspectives beyond our own. This is not an argument against conviction, but for recognising that even our strongest convictions must remain answerable to what we have yet to understand.

Ark seeks to give these ideas a practical expression. Its founding philosophy is intended to guide not only the institution's purpose, but its relationship with the people who depend on it. Policies and mechanisms are established to serve that purpose, not to be preserved for their own sake. Commitment to Ark should therefore mean remaining faithful to the search for truth, even when doing so challenges the decisions on which it was built.

3.2 Belief and participation

A philosophy acquires economic significance when people commit resources and work to what it makes possible. Those who believe in Ark's purpose can acquire NOAH, help secure the network, contribute to its development, and exercise influence over its governance. Their conviction can therefore help establish the economic and institutional foundations upon which broader participation becomes possible.

Participation, however, need not begin with shared philosophical conviction. People may use Ark's currencies to meet their economic needs, build services upon its infrastructure, or bond NOAH to earn rewards for securing the network. Their reasons for participating may be entirely practical, grounded in the value of what the institution provides rather than agreement with the ideas that inspired it.

Through participation, these people also have an opportunity to encounter Ark's philosophy in practice. Its principles become visible through the decisions it makes, the responsibilities it honours, and its willingness to learn from experience. Ark seeks to demonstrate their value through its conduct, strengthening confidence in the institution while inspiring people to embrace those principles for themselves and carry them into other aspects of their lives.

This relationship between belief and participation also matters to NOAH. Users exchange NOAH for newly issued currencies and receive NOAH through redemption, while bonding supports network security and carries governance authority. These functions create uses for the asset, but conversion activity alone does not ensure a lasting willingness to hold or bond it. That willingness depends in part on confidence in Ark's future, which the institution seeks to earn through both its economic usefulness and its continued commitment to its founding principles.

3.3 Principles in practice

Ark's founding principles take practical form in how the institution handles commitments and authority. Its capital policy assigns expansion proceeds to defined purposes, while its currency states specify how restrictions, settlement, and unpaid obligations are treated. These arrangements make the consequences of monetary decisions visible to the people affected by them.

Recognition of fallible judgement also shapes governance. Committees exercise limited authority for defined terms, their actions remain subject to public scrutiny, and governance can revise policies as circumstances and understanding change. The ability to correct a decision is part of the institution's design, alongside the authority to make it.

These responsibilities extend beyond those who exercise governance authority. People who hold and use Ark's currencies may depend on decisions made by NOAH holders without having a direct vote themselves. Governance must therefore remain accountable to everyone who depends on the system for the consequences of its decisions. Ark's founding philosophy provides a continuing standard against which those decisions, and the institution itself, can be judged.

4 Countercyclical capital

4.1 Capital retention and allocation

In ordinary conversion, users supply NOAH to create an Ark currency or return issued currency to receive NOAH. The protocol quotes these exchanges using the oracle value of the underlying fiat unit, with conversion charges applied.

Ark's capital policy begins from the dependence of its monetary commitments on NOAH's value and liquidity. Expansion should strengthen the institution's ability to manage the pressures that contraction can place on NOAH, while preserving resources for its monetary obligations. Ark therefore retains part of the NOAH received through currency creation in three funds with distinct purposes, giving priority to monetary sustainability over growth for its own sake.

The Redemption Buffer commits NOAH to automatic, proportional redemption funding, offsetting part of the new issuance required during contraction. The Strategic Reserve preserves resources for deliberate intervention, including commitments to the Buffer or Insurance, authorised burns, and deployments under a governance mandate. Insurance finances approved losses, supporting confidence in holding and using Ark's currencies. Keeping these functions separate ensures that resources committed to one purpose cannot simultaneously be used for another.

Expansion proceeds are allocated in order of priority: first to the Redemption Buffer, then the Strategic Reserve, and finally Insurance. Each fund is replenished up to its capital target before proceeds reach the next, with any remaining NOAH burned (Figure 1). This ensures that expansion first prepares the system for redemption and intervention before financing compensation for individual losses or reducing NOAH's supply. When proceeds are insufficient, later funds may receive little or nothing.

These arrangements provide Ark with different ways to respond to changing circumstances, but retaining NOAH does not itself create independent backing for its currencies. NOAH held by the funds remains exposed to changes in its market value, and valuing one NOAH at one NOAH for accounting purposes does not protect against a decline in purchasing power. Similarly, the Buffer's coverage ratio measures inventory available to offset new issuance, not the likelihood of maintaining a currency's peg. The funds are therefore reported separately rather than combined into a single measure of backing.

4.2 Currency liabilities

Ark's capital requirements depend on the obligations its currencies create. Because these currencies track fiat values but are redeemed through NOAH, the amount of NOAH required to meet those obligations changes with its market value. When NOAH declines, the same outstanding currency supply represents a larger liability in NOAH terms. The funds can therefore retain their balances while becoming less capable of supporting the commitments they were established to meet.

Ark distinguishes between gross liability, representing the total value of its recognised currency obligations, and net liability, which excludes currencies held by the Strategic Reserve. Net liability reflects the obligations held outside the Reserve and determines ordinary capital requirements and redemption funding. Gross liability is used to constrain discretionary intervention, since currencies held by the Reserve can subsequently return to

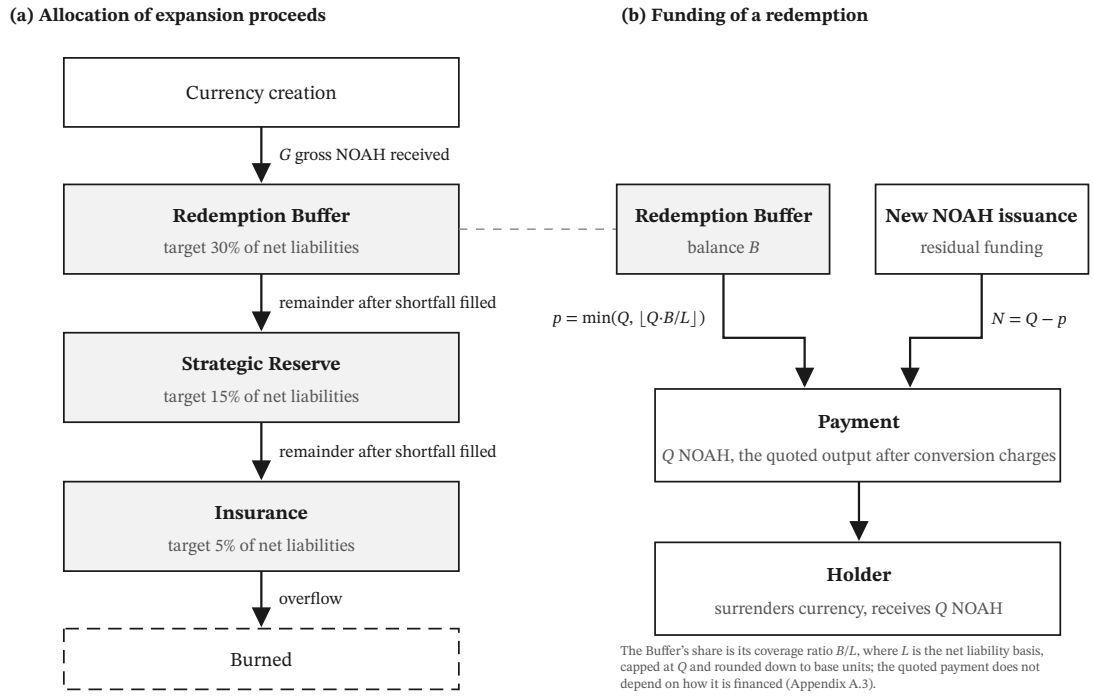


Figure 1. Allocation of expansion proceeds across the three funds (a) and funding of a redemption from the Buffer and new issuance (b), where the payment is the quoted conversion output after conversion charges and the Buffer's share is capped at that output and rounded to base units

circulation. This distinction allows Ark to account for its own holdings without allowing them to weaken the safeguards governing its resources.

These measurements must also account for uncertainty. Unavailable prices do not extinguish outstanding obligations, while settlement and write-off require explicit decisions about their treatment. Section 5 addresses these circumstances, and Appendix A.1 specifies the underlying accounting rules.

4.3 Capital targets and changing exposure

Ark establishes capital targets for each of its three funds based on the obligations its currencies create. At launch, the Redemption Buffer targets 30% of recognised liabilities, the Strategic Reserve 15%, and Insurance 5%. These targets express how much Ark seeks to retain for each purpose, rather than fixed shares of expansion proceeds or guarantees of coverage. As obligations grow, the resources required to meet those targets grow with them.

But the risks facing a monetary system do not depend on the size of its obligations alone. Changes in market conditions can make the same obligations increasingly difficult to sustain. Ark therefore includes an exposure multiplier designed to adjust capital requirements in response to the relative size of outstanding liabilities, volatility in NOAH's value, and sustained redemption pressure. When activated, the multiplier allows capital requirements to rise with exposure, directing more expansion proceeds towards the funds during periods of greater risk.

This adjustment is governed and remains inactive at launch, with its weights set to zero pending observation and calibration. Even when activated, higher targets cannot create

resources that do not exist. They can direct more proceeds from future expansion towards the funds, but cannot guarantee sufficient capital when demand contracts. The detailed calculations, adjustment limits, and launch parameters are provided in Appendices A.2 and B.2.

4.4 Redemption Buffer

The Redemption Buffer commits part of the NOAH received during expansion to future redemption. Its balance finances a proportional share of the NOAH paid to redeeming holders, with the remainder met through new issuance. The quoted payment is determined by the conversion rules; the Buffer determines how that payment is financed.

For example, if the Buffer covers 20% of outstanding net liability, it finances approximately 20% of a redemption, subject to rounding. Proportional funding distributes inventory across outstanding obligations, giving successive redemptions a share of the resources retained for them.

Retention reduces the amount of NOAH burned during expansion and subsequently issued during contraction. It establishes an explicit inventory commitment that can be observed alongside outstanding liabilities. Under the fixed-price comparison in Appendix A.3, retention and burn-and-reissue nevertheless produce the same flows to and from holders and finish with the same total NOAH supply. NOAH outside the Buffer also follows the same path.

NOAH released from the Buffer can be sold just as newly issued NOAH can. The Buffer's contribution is therefore a defined funding rule and committed inventory; any effect on confidence or market stability also depends on how participants respond to those arrangements.

Appendix A.3 specifies the allocation and redemption calculations and their execution timing, and Appendix A.4 establishes the conditional properties of proportional funding.

4.5 Strategic Reserve

Not every threat to monetary stability can be addressed through an automatic mechanism. The Strategic Reserve therefore preserves resources that Ark can deploy in response to changing circumstances. Unlike the Redemption Buffer, which is committed to ordinary redemption, the Reserve provides flexibility to respond to risks that require deliberate intervention.

These resources can be used to strengthen the Redemption Buffer or Insurance, reduce NOAH's supply, or acquire assets outside the monetary system. The ability to hold independently valued assets is particularly important because Ark's currencies ultimately depend on NOAH retaining sufficient market value. External assets provide a source of value beyond NOAH, although their inclusion introduces additional risks of custody, liquidity, and valuation.

Ark recognises external holdings conservatively, with each eligible holding subject to a valuation haircut and a cap tied to the Reserve's verifiable NOAH balance. Total recognised external capital cannot exceed that NOAH balance. Recognition does not establish liquidity or custody, and at launch only NOAH is recognised. Appendix A.5 gives the calculation.

Intervention also comes at a cost. Committing or deploying resources reduces what remains available for future needs, while losses or conservatively valued acquisitions may require replenishment from subsequent expansion. Decisions are therefore subject to governance and bounded delegated authority (see Appendix A.6 and Section 7). The Reserve cannot guarantee that an intervention will succeed, but it gives Ark the capacity to respond deliberately rather than relying exclusively on mechanisms established in advance.

4.6 Insurance fund

The risk of irreversible loss can make people reluctant to hold and use blockchain currencies. When a wallet or service is compromised, stolen funds may be difficult to recover. This sits within a broader problem of trust: a 2024 Pew Research Center survey found that 63% of US adults had little or no confidence in the safety and reliability of ways to invest in, trade, or use cryptocurrency [12].

Ark's Insurance fund provides resources for compensating approved losses. It begins with 5,000,000 NOAH and has an initial target of 5% of recognised net currency liabilities. Expansion replenishes it after the Buffer and Reserve meet their targets. The proposed coverage concerns theft losses involving registered members' issued currencies, rather than NOAH itself, under a published charter defining eligibility, evidence requirements, coverage limits, and the treatment of claims.

An appointed claims committee can submit covered claims within its authorised term allowance. Claims must be supported by available resources, and pending claims reserve funds that cannot simultaneously be treated as available for further commitments. Claims remain subject to cancellation before payment, with a one-week cancellation period at launch.

Linking coverage to verified identities associated with on-chain addresses gives participants a practical reason to establish a verifiable relationship with their holdings. Verification remains distinct from access to the monetary system: people can use Ark's currencies without seeking Insurance coverage. Identity-linked records may also assist investigation and accountability when theft occurs.

Insurance supports adoption by offering a defined form of protection to people who value it. Its capacity depends on the resources available, the quality of claims assessment, and the coverage commitments established in the charter. Future expansion may replenish the fund, but current claims must be managed against the capital actually available to meet them.

5 Currency distress and resolution

Ark's capital arrangements prepare the monetary system for contraction, while its safeguards and resolution procedures address disruptions to ordinary operation. Different circumstances require different responses, from temporary restrictions when reliable prices are unavailable to suspension, settlement, or write-off when monetary obligations can no longer be sustained. Defining these responses in advance clarifies who may act, how outstanding obligations are treated, and what each decision means for currency holders.

5.1 Oracle reliability and safeguards

Reliable pricing is essential to Ark's monetary system. Its currencies are issued and redeemed against fiat reference values expressed in NOAH, while capital requirements depend on the value of outstanding obligations. Unavailable or incorrect prices can therefore expose the system to losses through mispriced conversions and inaccurate capital accounting.

Each validator runs a price-feed service that observes external markets off-chain and reports the result in a consensus vote extension, so block execution never queries an exchange, and every node applies the same deterministic aggregation to the signed reports. A currency is priced only when validators holding more than two-thirds of commit power report a positive rate for it; validators with missing or invalid reports stay in the denominator, so silence cannot lower the threshold. Aggregation takes a voting-power-weighted median through a reference target chosen for the breadth of its reporting overlap and derives the other rates from weighted medians of the cross rates reported by validators who supplied both. A currency without adequate evidence receives no new price.

The oracle values the underlying fiat unit in NOAH, not the secondary-market price of Ark's corresponding currency. Conversion refers to that unit, subject to spreads and currency-state restrictions. A valid feed does not establish that the currency is maintaining its peg; secondary-market prices and conversion flows must be observed separately. Agreement among validators also cannot establish the accuracy of their sources, since a shared incorrect source can produce a consistent but wrong price.

Conversion paths close when their required prices fail availability or validity checks, while activity supported by usable prices may continue. These checks cannot detect every incorrect price. Following a network stall, proposers are expected to withhold user transactions until prices observed after the outage can be applied, although this precaution is not enforced by other validators.

When outstanding obligations cannot be valued reliably, Ark directs the block's gross expansion proceeds to the Reserve and burns none of them. Uncertain asset valuations are excluded from available capital. A suspended currency without a settlement plan is excluded from the valued-liability total, while its unpaid obligations remain disclosed. Writing off those obligations requires a separate governance decision. Appendix A.1 specifies the valuation rules for each currency state. These safeguards prevent the allocation of surplus against an incomplete valuation of the system's liabilities.

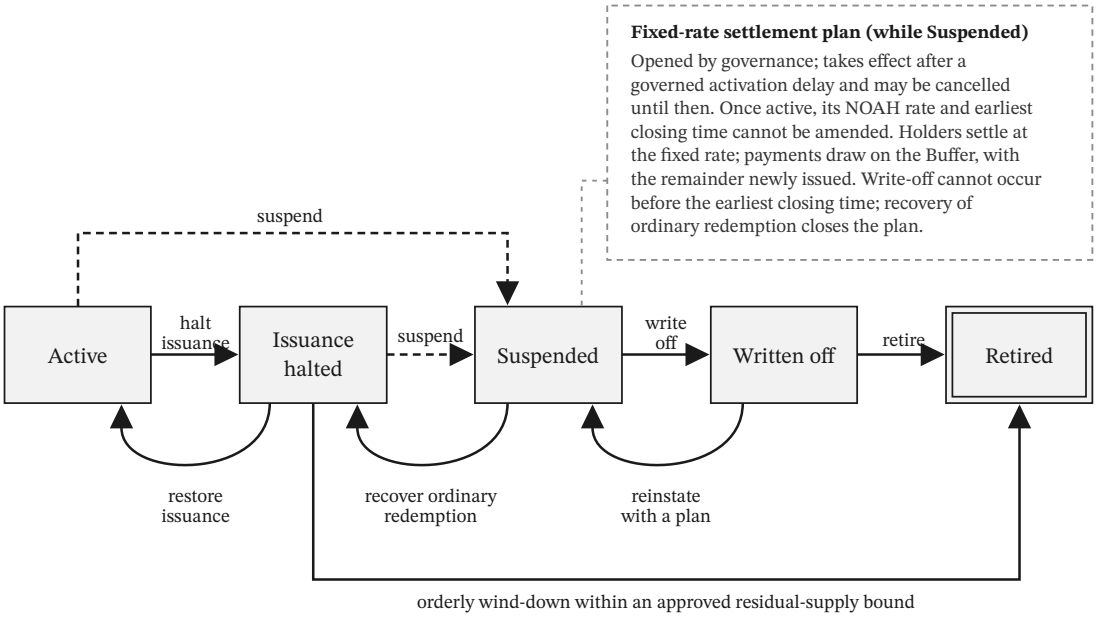
Missing information can be managed through temporary restrictions, but inconsistent financial records or invalid accounting cannot be treated in the same way. Appendix A.1 specifies liability treatment and Appendix B.2 lists the oracle settings.

5.2 Currency states

Ark's currencies share NOAH as their redemption asset. Heavy redemptions in one currency can increase the amount of NOAH offered for sale, affecting the value and liquidity on which the others depend. Different circumstances therefore require different responses. Further issuance may become inadvisable while existing redemptions remain sustainable; in more severe circumstances, continued redemption may threaten the wider monetary system. Ark establishes distinct currency states so that issuance and conversion can be restricted as conditions require, while preserving defined routes for recovery,

settlement, and the treatment of unpaid obligations.

Governance retains authority over currency status and resolution (Figure 2). The delegated asset emergency committee may suspend ordinary conversion, but cannot set settlement terms, write off obligations, or restore issuance. That division allows a faster response to immediate threats without granting emergency operators control over the eventual recovery or loss borne by holders.



Dashed arrows: governance, or the delegated Asset emergency committee (once per asset per appointment term).
Solid arrows: governance only. Double border: terminal state.

Figure 2. Currency state transitions and the authority required for each

Such interventions involve difficult trade-offs. Suspending conversion may protect the broader system from further deterioration, but it also restricts holders' ability to redeem their currencies and may expose them to losses. Ark therefore preserves their balances and ability to transfer them, rather than extinguishing their holdings through suspension. If conditions improve, ordinary conversion may be restored. Otherwise, governance must consider alternative settlement terms or acknowledge that the remaining obligations cannot be fulfilled.

The purpose of defining these states in advance is to establish a clear framework for responding to monetary distress. Rather than improvising the available actions during a crisis, Ark specifies the forms of intervention and the authority required to undertake them. This cannot guarantee timely intervention or prevent losses, but it provides a process through which difficult decisions can be made while remaining accountable to those affected by them. Table 1 summarises holders' positions in each state.

Table 1. Currency states and holder positions

State	Issuance and ordinary conversion	Holder position
Active	Currency can be issued, exchanged, and redeemed with usable prices	Ordinary conversion rights under current rules
Issuance halted	Currency can be consumed but cannot be newly produced	Redemption continues; issuance can later resume
Suspended	Ordinary conversion is closed	Balances transfer; an activated plan may provide fixed-rate settlement
Written off	Conversion is closed and the obligation is derecognised	Balances transfer; the loss and residual supply remain recorded
Retired	Terminal state; the denomination cannot be reused	Any permitted residual balances transfer without a protocol redemption obligation

5.3 Fixed-rate settlement

Suspending a currency prevents further issuance but leaves outstanding obligations to its holders. When ordinary redemption can no longer be sustained, governance may establish a fixed rate at which holders can exchange their currency for NOAH, potentially below its original reference value. With further issuance halted, the outstanding supply and settlement rate establish a maximum aggregate NOAH obligation. Settlement bypasses the ordinary oracle rate, virtual pool, and conversion spread, drawing proportionally from the Redemption Buffer, with any remainder financed through new NOAH issuance.

Settlement terms are determined by governance and take effect after a defined activation delay. A plan may be cancelled before activation, but its terms cannot be amended once active. Each plan states a settlement window; its earliest closing time is activation plus that window, fixed when the proposal is written, and no remaining obligation may be written off before it. If conditions improve, governance may restore ordinary redemption, replacing the settlement arrangement.

Although the obligation is bounded in NOAH, its full amount need not be prefunded, and NOAH's market value may change during settlement. Additional issuance or the release of existing NOAH into circulation may place further pressure on its price, affecting the value recovered by holders and potentially the stability of other currencies. Fixed-rate settlement therefore provides a defined means of resolving outstanding claims, while recognising that the economic consequences of those settlements extend beyond the obligations themselves.

5.4 Write-off and retirement

Not every monetary obligation can ultimately be fulfilled. When recovery or settlement is no longer viable, continuing to recognise a currency's outstanding obligations may create the appearance of commitments the institution can no longer sustain. Ark therefore provides a process for explicitly acknowledging such failures, rather than allowing unresolved obligations to remain indefinitely within the monetary system.

A write-off records the decision to cease recognising a currency's remaining obligations. It closes any settlement plan subject to its promised redemption window, while preserving

outstanding balances and the historical record of what remains unpaid. Unlike retirement, a write-off is not necessarily permanent. Governance may later reinstate the currency for recovery or settlement if circumstances permit, without implying that such recovery is guaranteed.

Retirement serves a different purpose. It permanently closes a denomination and prevents its reuse, preserving a clear record of its conclusion. A suspended currency cannot simply be retired with outstanding supply to avoid acknowledging its unresolved obligations. An orderly wind-down of an issuance-halted currency may, however, permit retirement within an explicitly approved limit on residual supply. These distinctions ensure that the system records which obligations were fulfilled, which remain unresolved, and which have been formally written off.

Neither write-off nor retirement restores value that holders have lost. Insurance may provide separate compensation for approved losses, but no currency resolution decision creates an automatic entitlement to those resources. The purpose of these procedures is not to make failure disappear, but to ensure that it is acknowledged transparently, that the consequences for holders are made clear, and that the institution can move forward without concealing the commitments it could not fulfil.

6 Network funding

Ark finances the continued operation of its validators and oracle participants through gas fees, transfer-tax revenue, and a dedicated subsidy pool. Operating finance is kept separate from the Redemption Buffer, Strategic Reserve, and Insurance, preserving those resources for redemption, intervention, and compensation. As economic activity develops, earned revenue is intended to support an increasing share of network operation, with the finite subsidy pool available to cover shortfalls.

6.1 Fees and transfer tax

Ark finances network operation through gas fees and a transfer tax on issued currencies. Gas fees compensate transaction processing, while the transfer tax generates revenue from currency use. These sources can support continued operation without requiring monetary expansion.

Ark uses XDR, the code for the IMF's Special Drawing Right (SDR), as its protocol reference unit at launch. The SDR's value is based on a basket of currencies.

Gas prices adjust according to network utilisation. At launch, the transfer tax is 0.5% of the value transferred, capped at the equivalent of 100 XDR per taxable transfer. NOAH transfers and ordinary conversions are not subject to this tax. Section 6.2 explains how gas-fee and transfer-tax revenue contribute to validator and oracle funding.

Conversion costs affect a different part of the user's transaction: the amount received when exchanging assets. Launch settings include a 2% minimum conversion spread and a 0.25% default charge for direct exchanges between issued currencies (see Appendix B.2). Together with gas and any applicable transfer tax, these costs determine the expense of using Ark's currencies and influence incentives to correct secondary-market price deviations.

Funding policy must balance compensation for those operating the network with the cost

imposed on users. Higher charges may increase revenue per transaction while discouraging activity; lower charges leave less revenue available from a given level of use.

6.2 Funding targets and shortfalls

Ark establishes separate funding targets for validators and oracle participants, reflecting their distinct responsibilities. Gas fees contribute directly to validator compensation, while transfer-tax revenue first supports the oracle funding target. Tax revenue beyond that requirement may contribute towards validator shortfalls, with any remainder also allocated to the oracle.

When earned revenue is insufficient, the subsidy pool provides existing NOAH to cover the remaining shortfalls. Funding is assessed over defined periods, allowing revenue collected during more active blocks to offset weaker periods before subsidies are used. If the subsidy cannot fully meet the outstanding targets, its available balance is distributed proportionally between them.

These targets represent intended compensation rather than guaranteed returns. Importantly, a funding shortfall does not trigger additional NOAH issuance or draw resources from the Redemption Buffer, Reserve, or Insurance. Ark instead acknowledges the shortfall, preserving the distinction between the resources available to operate the network and those committed to its monetary obligations.

Validators earn oracle rewards for reports within a governed band around the median, weighted by voting power and the number of targets rewarded, and a validator whose attendance over a weekly window falls below the governed minimum is jailed for a week rather than slashed. Appendix A.7 specifies the funding calculations.

6.3 Subsidies and long-term sustainability

During Ark's early stages, transaction activity and currency circulation are expected to be limited, providing little recurring revenue despite the need to maintain a functioning network. The subsidy pool therefore serves primarily to bootstrap validator and oracle compensation while the system develops and adoption grows.

The pool begins with 50,000,000 NOAH from the initial supply. Initial validator and oracle funding targets scale with the number of founding validator seats, establishing an operating budget before substantial transaction revenue develops. Additional validators do not automatically increase these targets; adjustments remain a governance decision. At ten launch seats and unchanged targets, the pool funds approximately 38.1 years of the NOAH budget with no revenue at all; that figure does not establish coverage of real operating costs.

As economic activity develops, earned revenue is intended to finance an increasing share of network operation, reducing reliance on the subsidy. The pool nevertheless remains available to cover subsequent shortfalls while resources remain, providing continuity through periods of weaker activity rather than ending automatically once the network matures.

The subsidy is finite, fee revenue depends on economic activity, and the purchasing power of NOAH-denominated rewards depends on NOAH's market value. A substantial token balance may still provide inadequate compensation if its value declines. Governance may

adjust funding targets, replenish the pool, or return some of its balance to the community pool. A withdrawal reduces the funds available for future shortfalls and must respect the minimum remaining balance stated in the proposal. These decisions must balance the resources consumed against the need to maintain sufficient participation.

Ark's objective is therefore to establish a path from subsidised operation towards funding sustained by ordinary economic activity, while remaining prepared for periods when that revenue proves insufficient. Currency use contributes to the compensation of validators and oracle participants, giving those operators an economic reason to maintain bonded stake and provide network security and reliable pricing. Separating operating finance from monetary capital preserves resources committed to currency holders while funding those who maintain the network. Launch allocations, funding parameters, and budget assumptions are documented in Appendices B.1 and B.2.

7 Participation and governance

7.1 Openness and neutrality

Ark's founding philosophy recognises the limits of any single understanding, including that of its founders. Its direction should therefore be shaped by its participants rather than permanently determined by those who established it. People should be able to use its currencies, question its policies, and contribute to its development without sharing the beliefs that inspired it.

Openness does not mean unrestricted participation. Currency users, member grant recipients, validators, and those exercising governance authority have different requirements and responsibilities. These distinctions should be transparent and justified by their practical purpose rather than ideological agreement.

Governance authority follows bonded NOAH, but its decisions affect a broader group, including currency holders who have no direct vote. Those exercising authority therefore have a responsibility to consider the interests of everyone who depends on the monetary system, not only NOAH holders. Ark's neutrality depends on maintaining this broader responsibility while allowing its rules and decisions to remain open to challenge and revision.

7.2 Currency use and fiat market integration

Ark's currencies are intended for ordinary economic activity: paying for goods and services, receiving income, and meeting commitments in familiar national denominations. The aim is to develop reasons to hold and use those currencies that persist beyond expectations of returns from cryptocurrency markets.

The intended experience requires services through which people can acquire an Ark currency, hold and transfer it in a wallet, pay a recipient, and exchange it when their needs change. Access from bank accounts and other fiat payment arrangements depends on services that connect those markets to Ark. Such integration connects the protocol's monetary functions with the financial relationships in which people already participate.

Ark disables Inter-Blockchain Communication (IBC) as part of this direction. This choice places less emphasis on connections to other cryptocurrency networks and seeks to

encourage development around fiat access and payment services. It may also narrow access to liquidity available through other networks.

The adoption strategy therefore depends on building useful routes into and out of Ark's currencies and reasons to use them between those points. Disabling IBC neither establishes those services nor prevents speculative trading; the practical work is to make Ark useful in the economic activities it is intended to support.

7.3 Ownership distribution

Ark begins with 1,000,000,000 NOAH, of which 782,000,000, or 78.2%, is reserved for distribution to members and contributors. The member pool holds 469,200,000 NOAH, while the contributor pool holds 312,800,000. This allocation reflects Ark's commitment to becoming a neutral monetary institution whose ownership and direction can extend beyond its founders. Rather than concentrating the initial supply among founding validators or outside investors, the vast majority is designated for broader participation. There is no separate investor, backer, or public allocation at launch (see Appendix B.1 for the complete genesis allocation).

Ark distributes these resources through three forms of grants, designed to extend ownership broadly while recognising the different contributions people make to the institution. Grants are committed in full when awarded and released according to predetermined schedules (see Appendix B.3). Unpaid grants carry no voting power; governance influence arises only as recipients receive NOAH and choose to bond it.

Table 2. Distribution grants

Grant	Purpose and authority
Member	Broadens ownership through grants from the member pool, awarded by the registrar within budgets approved by governance.
Ownership	Recognises contributions with a lasting stake in Ark through grants from the contributor pool approved by governance.
Compensation	Rewards ongoing work and specific contributions from the contributor pool, awarded by governance or an authorised grants committee.

Membership is established through an off-chain admission process intended to prevent duplicate registrations, with applicants vouched for by existing members. Governance limits the resources available for each round of registration, while delayed payments provide time to review or cancel improper awards. These safeguards seek to protect the member pool so that resources designated for broad distribution remain available to legitimate participants.

Ownership and compensation grants have separate limits intended to preserve resources for future contributors. These limits apply to beneficiary records and compensation rates, rather than a person's aggregate holdings across grants or other forms of distribution. Their scope must therefore be considered alongside the admission process and governance's assessment of individual awards.

Member distribution also proceeds in stages, allowing participation to develop as ownership spreads. A later stage opens only once independently bonded public stake

has reached a threshold set against earlier distributions, while contributor awards follow a staged allocation guideline. Both are governance policies rather than protocol rules, allowing the pace of distribution to respond to how the institution develops. Appendix B.3 records the stage plan and its thresholds.

The staged programme links further distributions to independent participation. Giving members and contributors an ownership stake is intended to create a continuing interest in the quality of Ark's services and a means of influencing their development. Its economic purpose is to encourage people to keep using, maintaining, and improving the institution in which they hold that stake. Recipients remain free to hold, sell, or bond their NOAH. Section 7.4 explains how these choices affect the distribution of governance authority.

7.4 Founding authority and governance transition

Ark cannot begin with the broad and independent participation it ultimately seeks to establish. During its early stages, the network requires validators to maintain consensus and a governing body capable of making decisions before a wider community has developed. Ark therefore begins under founding authority, with the intention that control progressively passes to participants as ownership is distributed and independently exercised.

Launch validators receive staking grants and an operating allocation. The founder's staking grant equals half the combined grants of the other launch validators, representing one third of the initial founding stake regardless of their number. These allocations provide the stake needed to establish the network and give its initial operators a continuing interest in its operation. The staking grants can be staked, voted, and slashed from launch, but become transferable only through vesting between the fourth and eighth years.

This arrangement gives the founder a significant role in Ark's initial governance. Under the launch voting rules, the founder's share of founding stake is sufficient to block ordinary proposals, but insufficient to approve them alone. This is intended to provide continuity while the network's monetary and institutional arrangements are established. The blocking position arises from ownership rather than a permanent privilege, and can diminish as independent participation develops. Appendix B.4 gives the voting rules and an illustrative calculation.

The founder also receives an ownership grant of 15,000,000 NOAH for the work undertaken to establish Ark and a compensation award of 10,000,000 NOAH, paid over ten years, for continued development and maintenance. Both awards are deliberately set at their respective limits under the initial contributor framework and follow the same payment schedules and cancellation rules as other contributor grants. As with those grants, unpaid awards carry no voting power; authority arises as NOAH is paid and bonded. Appendices B.1 and B.3 give the complete allocation and vesting rules.

Ark also distinguishes governance authority from control over consensus. The founder's excess staking allocation is delegated across the other launch validators rather than concentrated behind a single node. This distributes consensus voting power without transferring ownership of the delegated stake or the governance authority associated with it. Decentralising network operation does not necessarily decentralise the decisions governing the network.

Following launch, validator participation is open to those who bond sufficient NOAH,

without requiring a founding allocation or governance appointment. As members and contributors receive and bond their grants, they can exercise governance authority directly or through delegation. The founder's blocking position can diminish as independently exercised public stake grows.

The distribution of authority depends on how that stake is held and used. Recipients may sell, leave their NOAH unbonded, or delegate to existing validators, and control can become concentrated again after it has broadened. The transition therefore rests on continuing independent participation as well as the initial distribution of ownership.

7.5 Governance and delegated authority

Monetary policy, Reserve management, and emergency response require decisions at different speeds and levels of detail. Governance can delegate defined responsibilities to appointed committees so that routine administration and urgent action can proceed within an agreed mandate.

Governance determines each committee's scope, appoints its members, and can replace or withdraw its mandate. Table 3 summarises the delegated roles.

Table 3. Committee roles and authority

Role	Delegated authority
Economic policy	Adjust reward targets, fund ratios, and exposure weights within a corridor
Conversion	Adjust pool policy and Tobin charges within a corridor
Reserve	Deploy, record positions, commit funds, and burn within specific bounds
Claims	Submit covered claims within a term allowance and cancel eligible committee claims
Asset emergency	Suspend an eligible currency once per currency per term
Security	Schedule or cancel its own upgrade plan
Grants	Appoint the registrar, suspend and reinstate members, award compensation within a term allowance

Committee mandates last for defined terms and require reappointment. Governance retains control over those mandates and over decisions not expressly delegated, including changes to the oracle's reference unit and price feeds. The chain begins without appointed committees; their authority takes effect through subsequent governance decisions. Appendix B.5 records the intended appointment triggers and initial limits, while Section 7.6 explains the safeguards governing their actions.

7.6 Safeguards and accountability

Ark constrains delegated authority through policy corridors, spending limits, resource requirements, and restrictions on the actions each committee may take.

Committee adjustments to monetary policy are restricted to ranges established by governance, while committee deployment of Reserve resources is constrained by spending allowances, approved destinations, and a minimum NOAH balance, and committee burns are further bounded by the Reserve's capital target. Insurance claims must be supported by

available resources and remain subject to cancellation before payment. These safeguards prevent committees from making commitments beyond their authorised limits, while ensuring that existing obligations cannot be circumvented through changes in custody, appointments, or accounting.

Full governance can authorise Reserve transfers and burns without the target and allowance bounds imposed on committees. These actions must still satisfy the minimum balance specified in the proposal and the other applicable validation rules. Appendix A.6 specifies the distinction between governance authority and the financial bounds on Reserve committee actions.

Emergency powers require particular care. When a currency comes under severe stress, delayed intervention may allow further deterioration, but acting too broadly could impose unnecessary losses on holders. Ark therefore allows the asset emergency committee to suspend ordinary conversion without granting authority to determine settlement terms, extinguish obligations, or confiscate balances. Similarly, the security committee can schedule or cancel its own software upgrade plans without acquiring general control over monetary policy or protocol funds. These distinctions allow immediate protective action while reserving more consequential decisions for governance.

Effective intervention also depends on the network's ability to process authorised actions during periods of stress. Exceptional transaction volume could otherwise delay the very decisions needed to protect the monetary system. Ark therefore reserves mempool capacity and gives eligible governance and committee transactions preferential treatment, reducing the risk that ordinary congestion prevents timely intervention. This cannot guarantee inclusion against validator censorship or network failure, but it improves the institution's ability to act when intervention is most needed.

Public records of appointments, policy changes, financial commitments, and resolution decisions allow participants to examine how authority has been exercised. They provide the basis for challenging decisions, replacing delegated authorities, and revising policies. Governance retains the ability to change its own safeguards, making scrutiny of those decisions part of the institution's continuing accountability.

8 Future development

Ark is designed to develop progressively rather than require every proposed service to be available from its first block. The initial network establishes NOAH, its monetary infrastructure, governance arrangements, and operating funds. Currency issuance, member distribution, and Insurance coverage can subsequently be activated as their respective requirements are met. Governance opens currency issuance by registering the currencies once NOAH has a market price. Member admission and distribution require supporting applications and identity arrangements, while Insurance requires a coverage charter before protection can be offered.

Ark's monetary arrangements will continue to evolve through governance. Exposure adjustment begins with zero weights pending calibration, while the launch parameters remain subject to revision (see Appendix B). The published source repository at the cited commit and the final genesis configuration will provide a reproducible basis for examining the implementation and its subsequent development [13].

Ark's first currencies track familiar national monetary units, allowing participants to hold and exchange value in denominations they already understand. These currencies provide a foundation for economic activity across jurisdictions, while establishing the infrastructure and relationships from which a broader monetary system can emerge.

Orbis extends this vision towards a neutral currency independent of any national monetary unit or fixed basket of currencies. Unlike Ark's fiat-tracking currencies, whose values are defined by external reference units, Orbis would establish its own monetary identity through the economic relationships and demand that develop around it. Ark's existing currencies could provide the means through which people acquire, exchange, and begin using Orbis, allowing a shared currency to emerge alongside the national denominations already supported by the network.

Orbis requires a distinct monetary design. Adopting the issuance and redemption rules of Ark's fiat-tracking currencies would preserve their dependence on an external reference value. Its supply policy, relationship to NOAH, and mechanisms for sustaining monetary confidence will therefore require separate development and governance. The ambition is to establish a currency whose value is sustained by its own economic use rather than a fixed external denomination.

9 Conclusion

Ark seeks to build a monetary institution in which economic usefulness and shared purpose reinforce one another. Its founding philosophy offers a basis for confidence in the institution behind NOAH, while its currencies and services give people practical reasons to participate. Belief can motivate commitment, but lasting confidence must be earned through the institution's conduct and its ability to serve those who depend on it.

That responsibility is reflected in Ark's monetary design. It retains resources against defined commitments, establishes how monetary distress and unpaid obligations are treated, and provides a path towards shared ownership and governance. These arrangements make stability an ongoing institutional responsibility, but they cannot guarantee it. Ark's currencies remain dependent on NOAH's market value and liquidity, and the institution must remain willing to question and revise its decisions as experience reveals their limitations.

Ark begins with a philosophy, but participation need not depend on agreement with it. People who come to Ark for its economic benefits will also encounter its principles through the way the institution operates. By demonstrating their value in practice, Ark seeks to deepen confidence in the institution while inspiring people to carry those principles into other aspects of their lives. As participation broadens, its future should increasingly belong to the people who use, sustain, and shape it, united not by identical beliefs, but by their commitment to what they build together.

Appendix A Monetary calculations

These relationships summarise the main text in economic notation; they are not a second normative specification. The implementation remains authoritative for integer arithmetic, price eligibility, and execution order. Floors and ceilings round to whole NOAH base units.

A.1 Currency liabilities and capital targets

For each included currency i, let s_i be outstanding supply, h_i the supply held by the Reserve, and r_i its value in NOAH per unit. Gross and net recognised liabilities are L_g and L_n, both measured in NOAH.

Each included currency uses the rate assigned by its pricing branch: fresh oracle, an opened settlement plan, or separately identified last-known evidence. Unvalued exposures remain disclosure entries rather than terms with fabricated rates.

Table 4. Valuation by currency state

State	Rate	Counted in liability	Ordinary allocation
Active or issuance halted, fresh oracle rate	Oracle rate	Yes	Yes
Active or issuance halted, feed unavailable, previously priced	Last known rate, disclosed as stale	Yes	No: proceeds park in the Reserve
Active or issuance halted, never priced	None	No, disclosed	No
Suspended under a settlement plan, from its opening	Plan rate, before and after activation	Yes	Yes
Suspended without a plan	None	No, disclosed as untrusted	No
Written off	None	No, disclosed	Yes
Retired	None	Not recognised	Yes

The redemption draw in Appendix A.3 divides by the net liability counted above plus the value the block retired, so supply that leaves the denominator lets the same redemption draw a larger share of the Buffer; stale and plan-rate supply keep their share.

For fund f, a_f is its target ratio, T_f its target amount, C_f its recognised resources, and gap_f its shortfall. Amounts are in NOAH; m is the bounded exposure multiplier defined in Appendix A.2.

$$L_g = \sum_i s_i r_i$$

$$L_n = L_g - \sum_i h_i r_i$$

$$T_f = \left\lceil a_f \, m \, \text{basis} \right\rceil$$

$$\text{gap}_f = \max(T_f - C_f, \, 0)$$

$$\text{basis} = L_n \quad \text{for ordinary expansion allocation}$$

$$\text{basis} = L_g \quad \text{for target-based committee bounds}$$

Incomplete valuation disables the ordinary target-based expansion allocation. Published targets are zeroed with explicit disclosure rather than presented as valid partial requirements; actual balances are unaffected.

A.2 Exposure multiplier

Let r be the reference rate in NOAH per unit, t the observation time, and r_0 and t_0 the sampling anchor and its time. The elapsed interval is $\Delta t = t - t_0$, measured in seconds.

In the update below, x is the bounded reference return, v the exponentially weighted mean of squared returns, a zero-mean variance estimate, τ the mean sample interval in seconds, and d_v the retention factor per volatility sample. When Δt reaches the sample spacing and a fresh rate is available, the series updates as follows:

$$x = \text{clamp}\left(\frac{r}{r_0} - 1, \, -1, \, 1\right)$$

$$v \leftarrow d_v \, v + (1 - d_v) \, x^2$$

$$\tau \leftarrow d_v \, \tau + (1 - d_v) \, \Delta t$$

$$r_0, \, t_0 \leftarrow r, \, t$$

Otherwise the anchor holds; an unavailable rate does not advance it. The first sample after genesis sets the anchor and yields no return. A change of reference rescales the anchor into the new unit where the rates allow, and otherwise clears it so that the next sample re-anchors.

For each block, let E be the NOAH value of liability added by expansions and R the liability retired by redemptions, also in NOAH. Net redemption pressure is P, measured in NOAH, and d_P is the retention factor per block:

$$P \leftarrow \max\left(0, \, d_P \, P + (1 - d_P)(R - E)\right)$$

Outside-fund NOAH, N_c, is total supply less balances in the subsidy pool, Buffer, Reserve, and Insurance. It includes the Disbursement pools and any other balances not excluded by that definition, so it is not a measure of freely tradable supply.

At each scheduled refresh, ρ is the liability ratio, σ annualised volatility, and φ the flow ratio. Their economic-policy weights are w_ρ, w_σ, and w_φ. Annualisation uses $Y = 31{,}536{,}000$ seconds per year.

The uncapped composite is $\hat{m}$; m_{prev} is the previous multiplier, $m_{\max}$ its cap, and δ the maximum change per refresh. The multiplier combines the latest available measures as follows:

$$\rho = \frac{L_n}{N_c}$$

$$\sigma = 0 \quad \text{(before the first return)}$$

$$\sigma = \sqrt{\frac{v\,Y}{\tau}} \quad \text{(otherwise)}$$

$$\varphi = 0 \quad (\text{when } L_n = 0)$$

$$\varphi = \frac{P}{L_n} \quad \text{(otherwise)}$$

$$\hat{m} = (1 + w_\rho\,\rho)(1 + w_\sigma\,\sigma)(1 + w_\varphi\,\varphi)$$

$$m = \min\left(m_{\max},\ \max\left(1,\ m_{\text{prev}} + \text{clamp}(\hat{m} - m_{\text{prev}},\ -\delta,\ \delta)\right)\right)$$

The targets defined in Appendix A.1 use m on both liability bases. The liability ratio ρ and the redemption draw in Appendix A.3 use unscaled figures.

The multiplier moves by at most δ per refresh, so reaching $m_{\max}$ from one requires at least $(m_{\max} - 1)/\delta$ refreshes, except where a lowered cap takes it further down in one refresh. The cap bounds the result after the step: a reduced cap limits the multiplier to the new cap, and the ordinary step may already have placed it lower.

A retention factor d has a half-life of $(\ln 2)/(-\ln d)$ samples. At the launch settings in Appendix B.2, the volatility half-life is about 23 hours at five-minute sampling, and the redemption-pressure half-life is about one hour.

Interpretation and limits

The multiplicative form allows each weighted indicator to contribute independently and compound with the others. A zero weight contributes one. All indicator weights are zero at launch. The multiplier is floored at one, capped by governance, and step-limited in both directions. With zero weights, a previously raised multiplier returns towards one by the permitted step rather than resetting immediately.

These choices bound and smooth the response; they do not prove resistance to all economically profitable manipulation. Conversion costs constrain some deliberate flow generation, while reference rates can also be affected by upstream market quality

and common provider dependence. The chain does not observe secondary-market peg deviation through the fiat-reference oracle. Reporting dispersion is not used as a substitute for that missing measure. The launch cap allows the three targets to sum to the whole liability basis; it does not establish that those resources are available.

If a refresh cannot obtain its required inputs, including a fresh reference or positive N_c, it retains the previous multiplier until the inputs are available. Partial liability valuations can still feed the indicator when its required inputs exist; excluded exposure is disclosed rather than assigned a fabricated value. The indicator can therefore understate unvalued exposure even while the separate incomplete-valuation rules preserve expansion proceeds.

A.3 Conversion flows and NOAH supply

Let G be gross NOAH received in the block's expansions and credit_f the amount credited to fund f. The fund shortfalls gap_f are defined in Appendix A.1; unallocated expansion proceeds are the overflow. The Reserve's credit is bounded not by its nominal shortfall but by the smallest amount whose post-credit recognised capital (Appendix A.5) reaches its target; the two coincide under NOAH-only custody.

For the block's redemptions, R is liability retired and Q the output paid, both in NOAH. The redemption basis is $L = L_n + R$, using post-redemption L_n. Let B be Buffer NOAH after expansion credits and before the draw, p the Buffer contribution, and N the residual net redemption issuance. When the block pays no redemption output, the Buffer contribution is zero and no quotient is formed. In execution, each redeemer receives the full quoted output when the transaction runs, and the block's Buffer contribution is transferred and burned at its end, so final supply and balances match the identities below.

$$\text{credit}_{\text{Buffer}} = \min(G, \text{gap}_{\text{Buffer}})$$

$$\text{credit}_{\text{Reserve}} = \min(G - \text{credit}_{\text{Buffer}}, \text{need}_{\text{Reserve}})$$

$$\text{need}_{\text{Reserve}} = \min\{x \geq 0 : C_R(B_R + x) \geq T_R\}$$

$$\text{credit}_{\text{Insurance}} = \min(G - \text{credit}_{\text{Buffer}} - \text{credit}_{\text{Reserve}}, \text{gap}_{\text{Insurance}})$$

$$\text{overflow} = G - \text{credit}_{\text{Buffer}} - \text{credit}_{\text{Reserve}} - \text{credit}_{\text{Insurance}}$$

$$p = \min\left(Q, \left\lfloor \frac{Q\,B}{L} \right\rfloor\right)$$

$$N = Q - p$$

$$\text{change in total NOAH from conversion} = N - \text{overflow}$$

The supply identity excludes separate Reserve burns and any change to protocol rules. During incomplete valuation, $\text{credit}_{\text{Reserve}} = G$, the other expansion credits and overflow are zero, and the draw still uses the available recognised net basis. Direct issued-currency exchanges create no NOAH.

Buffer funding and supply comparison

For example, with a liability of 1,000 NOAH and a Buffer holding 200 NOAH, coverage is 20%. A redemption quoted to pay 98 NOAH uses 19.6 NOAH from the Buffer and requires 78.4 NOAH of net new issuance, before base-unit rounding. The quoted output is an input to this calculation and is not determined by the Buffer's funding rule.

The difference between retention and conventional burn-and-reissue financing can be illustrated under fixed prices. Suppose total NOAH supply begins at 1,000, currency creation receives 100 NOAH, and the resulting liability is subsequently redeemed for 100 NOAH.

Table 5. Buffer retention and burn-and-reissue comparison

Stage	Burn and reissue	Retain 30% in Buffer
Expansion receives 100 NOAH	Burn 100	Retain 30; burn 70
Redemption pays 100 NOAH	Issue 100	Release 30; issue 70
NOAH delivered to redeemers	100	100
Total NOAH supply	1,000 → 900 → 1,000	1,000 → 930 → 1,000
NOAH outside the Buffer	1,000 → 900 → 1,000	1,000 → 900 → 1,000

The comparison assumes no initial Buffer, no other funds or interventions, no conversion charges, fixed prices, and complete redemption of the liability created during expansion.

Retention produces a smaller contraction and subsequent expansion of total NOAH supply. Under these assumptions, however, both approaches remove the same quantity from holders during expansion and deliver the same quantity during redemption, so NOAH outside the Buffer follows the same path. Retention makes the funding of future redemption explicit through committed inventory.

Any difference in market expectations or price volatility depends on participant behaviour, liquidity, and the credibility of those commitments. Economic comparisons must model these effects rather than assume newly issued NOAH creates more selling pressure than NOAH released from inventory.

A.4 Buffer coverage during redemption

Using the Buffer and redemption quantities from Appendix A.3, consider only redemptions, with fixed rates and liability membership, no other balance movements, $0 \le B < L$, and $0 < R < L$. Valid outputs satisfy $Q \le R$. Before base-unit rounding, let the coverage ratio be $c = B/L$ and the Buffer contribution $p = c\,Q$. Then:

$$\frac{B-p}{L-R} - \frac{B}{L} = c\,\frac{R-Q}{L-R} \ge 0$$

In exact arithmetic, flooring the contribution retains additional Buffer inventory. When $B \geq L$, the contribution is capped at the actual output; excess inventory is not awarded to the last redeemer. At $R = L$, there is no remaining liability against which to define a coverage ratio.

The proof does not cover price revaluation, changes in which liabilities are counted, expansion, or discretionary movements. It describes the proportional mechanism rather than a bound on secondary-market outcomes. In execution, the fixed-decimal quotient followed by integer truncation can differ from the exact rational floor by one base unit near a rounding boundary. The implementation retains exact custody and output bounds: $0 \leq p \leq \min(B, Q)$.

A.5 Recognition of external Reserve assets

Let B_R be Reserve NOAH. For each eligible holding j, u_j is the quantity attested within its window, r_j the rate in NOAH per unit, d_j its haircut factor, and k_j its maximum recognised credit per NOAH held by the Reserve. The holding's recognised value is credit_j, and total recognised Reserve capital is C_R, both in NOAH.

Rates must satisfy the holding's freshness policy; ineligible or unusable evidence earns zero credit.

$$\text{credit}_j = \min(d_j\, u_j\, r_j,\ k_j\, B_R)$$

$$C_R = B_R + \sum_j \text{credit}_j$$

$$\sum_j k_j \leq 1$$

$$C_R \leq B_R \left(1 + \sum_j k_j\right) \leq 2\, B_R$$

The cap restricts recognised credit even if reported quantities are arbitrarily large within their allowed domain, and each credit depends only on its own holding and B_R, so one misreported holding cannot raise another's. It cannot prove custody or ensure liquidation proceeds. Ark-issued currency contributes no such credit.

A cap based on recognised capital would allow one credit to enlarge the base used to recognise another. Measuring each cap against B_R avoids that feedback. A fixed nominal credit cap or one based only on target liability would produce different scaling and custody relationships. The chosen rule guarantees the algebraic bound above, not that NOAH is a risk-free asset or that external assets have been independently verified. Expansion credits the Reserve the smallest amount whose post-credit recognised capital reaches its target rather than the nominal shortfall: while a holding's cap binds, each NOAH credited recognises more than one, so the shortfall would carry the fund past its target and that excess out of Insurance and the burn.

A.6 Limits on Reserve committee actions

Let $C_R(b)$ be the Reserve's recognised capital at a NOAH balance b, using the holding quantities, rates, haircuts, and caps defined in Appendix A.5:

$$C_R(b) = b + \sum_j \min(d_j\, u_j\, r_j,\; k_j\, b)$$

Here B is the current Reserve NOAH balance, $b_{\min}$ the mandate's minimum NOAH balance, and A its remaining gross deployment allowance for the term, also in NOAH. Let T_R be the Reserve's gross-basis target and gap_f the gross-basis shortfall of the receiving fund (Buffer or Insurance). An action moves z NOAH; z^* is the maximum surplus burn allowed by the capital target and balance floor.

$$\text{deployment: } z \le A,\; B - z \ge b_{\min};\; \text{then } A \leftarrow A - z$$

$$\text{commitment: } z \le \text{gap}_f,\; B - z \ge b_{\min},$$

$$\text{for } f \in \{\text{Buffer, Insurance}\}$$

$$\text{surplus burn: } z \le z^*, \text{ where}$$

$$z^* = \max\{z \,:\, C_R(B - z) \ge T_R \wedge B - z \ge b_{\min}\}$$

$C_R(b)$ is non-decreasing in b, so z^* can be found by search on the post-burn balance. For a holding whose credit remains capped, recognised capital falls by $1 + \sum_{\text{capped}} k_j$ per unit burned, at most 2 when the caps sum to one. The pre-burn surplus is therefore only an upper bound on the permissible burn. With $B = 1{,}000$, $T_R = 1{,}200$, and one capped holding at $k = 0.3$, $C_R(1{,}000) = 1{,}300$ but $1.3\,(1{,}000 - z) \ge 1{,}200$ permits only about 76.923 units before rounding. In an integer-unit illustration the largest burn is 76; actual NOAH execution uses base-unit precision. The bound is zero when no non-negative burn satisfies every constraint, and in particular once $C_R(B) \le T_R$.

For a newly acquired holding j and other holdings that remain in the indicated capped regime throughout the movement, a deployment of z NOAH attested afterwards at value V in NOAH changes recognised capital by:

$$\Delta C_R = -z + \min(d_j\, V,\; k_j\,(B - z)) - \sum_{\substack{i \ne j \\ \text{capped}}} k_i\, z$$

With $V = z$, no other holding capped, and j 's cap slack, recognised capital falls by $(1 - d_j)\, z$. For a Reserve that stood at its target, that is the gap that reopens.

Governance transfers and burns do not use these committee target and allowance bounds. Their stated custody condition is that $B - z$ remain at least the minimum balance specified in the proposal, subject to the message's other validation rules.

Commitments to the Buffer and Insurance can proceed during incomplete liability valuation against the available valued basis; committee surplus burns cannot.

A.7 Validator and oracle funding

For one funding window, let V and O be the accumulated validator and oracle targets, F eligible gas-fee value, and T priced transfer-tax value, all measured in NOAH.

$$\text{validator gas gap} = \max(V - F,\ 0)$$

$$\text{protected oracle tax} = \min(T,\ O)$$

$$\text{desired validator tax} = \min(T - \text{protected oracle tax},\ \text{validator gas gap})$$

The desired validator fraction is applied to each tax denomination, rounding its allocation down. The oracle receives the remainder. Let U_V and U_O be the remaining validator and oracle shortfalls, revalued in NOAH after the actual coin allocations, and S the available subsidy NOAH. A sufficient subsidy pays both; otherwise:

$$\text{oracle subsidy} = \left\lfloor \frac{S\,U_O}{U_V + U_O} \right\rfloor$$

$$\text{validator subsidy} = S - \text{oracle subsidy}$$

Consider a window with a validator target of 100 NOAH and an oracle target of 20. Gas contributes 70 of recognised value and tax contributes 30. Tax allocates 20 to the oracle and 10 to validators, leaving a validator shortfall of 20 for subsidy. With only 8 NOAH of subsidy, validators receive that 8 and the remaining 12 is unfunded. The shortfall is an observable result, not an instruction to create the missing money.

No shortfall requires no subsidy. Let v and o be the constant validator and oracle targets in NOAH per block. With $v + o > 0$, zero qualifying revenue, and no replenishment or withdrawals, $S/(v + o)$ is the approximate funded number of blocks, subject to window settlement and rounding.

Appendix B Launch parameters and policies

This appendix records the launch allocation, the distribution and governance policies the main text relies on, and the settings it cites. The maintained launch record and curated genesis configuration hold the complete configuration and its assembly procedure [13]; this paper describes the repository at commit aee81ea. Block-to-time translations assume six-second blocks.

B.1 Initial supply and allocations

Table 6. Initial supply and allocations

Area	Launch value
Initial supply	1,000,000,000 NOAH
Subsidy pool seed	50,000,000 NOAH; 5% of initial supply
Reserve seed	50,000,000 NOAH; 5%
Buffer seed	10,000,000 NOAH; 1%
Insurance seed	5,000,000 NOAH; 0.5%
Disbursement pools	469,200,000 NOAH for members and 312,800,000 NOAH for contributors
Community pool	103,000,000 NOAH before launch seat allocations; the seat-count formula below determines the remainder
Seat staking grant	5,000,000 NOAH per seat, the founder's is half the others' sum, 67,500,000 at ten; one seat self-delegated at genesis, vesting from year four to year eight
Seat liquid float	50,000 NOAH per seat for operating expenses

For n launch validator seats, where $7 \leq n \leq 13$, the community pool retains $103{,}000{,}000 - 5{,}050{,}000\,n - 2{,}500{,}000\,(n-3)$ NOAH. The final term funds the founder's staking allocation above a standard seat grant. Seat allocations transfer existing supply; total initial supply remains 1,000,000,000 NOAH.

For ten seats, the allocations are 67,500,000 NOAH in seat grants, including the founder's 22,500,000, plus 500,000 liquid NOAH and 35,000,000 NOAH remaining in the community pool, alongside the four fund seeds and the Disbursement pools. This illustrates the per-seat rule; it is not an announced initial seat count.

B.2 Launch settings used in the text

The settings below are the ones the main text and Appendix A cite. Every other launch setting, among them the conversion pool depth and recovery, the oracle windows, the gas and base-fee parameters, the consensus and slashing parameters, and the governance deposits and periods, is recorded with its rationale in the launch record [13].

Table 7. Launch settings the text relies on

Area	Launch value
Currency family at opening	AUD, CAD, CNY, EUR, GBP, JPY, KRW, MXN, SGD, USD, registered together by governance when conversion opens
Routine NOAH issuance	None; no mint module or staking inflation
Buffer, Reserve, Insurance target ratios	30%, 15%, 5%
Exposure weights	All zero; initial multiplier one

Area	Launch value
Exposure multiplier cap and maximum step	2 and 0.1 per update
Exposure refresh	600 blocks, approximately one hour
Volatility sample spacing and decay	300 seconds of block time; 0.9975 per sample
Flow decay	0.99885 per block
Minimum conversion spread and direct currency conversion charge	2%; 0.25% default Tobin rate, no overrides
Transfer-tax rate and cap	0.5%; 100 XDR per taxable transfer
Claim cancellation period and settlement activation delay	604,800 seconds of block time each, one week
Oracle price quorum	0.666666666666666667 of total commit power
Ordinary price freshness	60 seconds; also the price age past which proposers leave user transactions out
Oracle reward band	2% total width around the relevant median
Oracle attendance window and jail	100,800 blocks, approximately one week; jailed for 604,800 seconds of block time, one week
Validator and oracle funding targets	0.0175 and 0.0075 NOAH per block per seat
Reward funding window	100,800 blocks, approximately one week
Initial NOAH fee-conversion factor	1 NOAH per XDR until the first usable oracle observation; a bootstrap assumption, not a market rate
Governance quorum, approval, and veto thresholds	50% quorum; 66.7% ordinary and 75% expedited approval; 33.4% veto

B.3 Distribution and vesting

Table 8. Distribution and vesting settings

Area	Launch value
Member grant	20,000 NOAH: a tenth after a two-week hold, then thirteen payments four weeks apart
Member tranche	Zero; each step opens by vote
Ownership schedule	A quarter after 364 days, then 78 fortnightly payments
Ownership cap	15,000,000 NOAH committed per record, checked at award
Compensation cap, window, and period	1,000,000 reference units per record per 26 fortnights; fortnightly, at most 260 periods
Founder's awards	15,000,000 NOAH of ownership and 10,000,000 NOAH of compensation over 260 fortnights, governance grants 1 and 2
Grants committee and registrar	Not appointed

Every grant's first period is at least two weeks, so nothing pays before two weeks after an award.

Compensation is valued in the reference unit at the oracle rates current when it is awarded, rounded up. The founder's stream is valued at the launch bootstrap rate of one reference unit per NOAH, so 10,000,000 NOAH over 260 fortnights charges exactly the 1,000,000 cap per 26-fortnight window, checked when seat assembly writes the award and again at every later award to the same record. Only governance cancels a grant; the grants committee and the registrar cannot, though the committee may suspend a member's payments. Cancellation pays what the schedule has earned but not yet paid and returns the unearned remainder to its pool or tranche; a cancelled compensation award still counts against the cap for its first window.

Staged distribution policy

The first member stage allocates 30,000,000 NOAH to 1,500 members, with subsequent stages progressively reducing individual grants to 5,000 NOAH by the fifth stage, reaching approximately 66,300 members in total.

Table 9. Distribution stages

Stage	Distribution budget, cumulative	Member pool, cumulative	Member grant	Members, cumulative	Contributor pool, cumulative	Gate
1	50,000,000	30,000,000	20,000	1,500	20,000,000	None
2	150,000,000	90,000,000	15,000	5,500	60,000,000	3,750,000
3	300,000,000	180,000,000	10,000	14,500	120,000,000	18,750,000
4	450,000,000	270,000,000	7,500	26,500	180,000,000	41,250,000
5	782,000,000	469,200,000	5,000	about 66,300	312,800,000	63,750,000

The table's figures are stage budgets, not the sums actually awarded. Each later stage opens by governance vote once independently bonded public stake reaches 15% of the cumulative distribution through the previous stage, counting the budgets opened whether or not every grant has been awarded or paid, net of the founder's 25,000,000 NOAH of contributor awards. That stake is the bonded total less the founding accounts' delegations, their seats, and whatever of their income and of the founder's paid awards they have bonded; paid awards left unbonded are not subtracted, so the figure counts only stake that is actually bonded and cannot fall below zero. The founder's awards are charged to the first stage's 20,000,000 NOAH contributor budget as an overrun the guideline permits, and they leave the gate base, so the first gate is 15% of the 25,000,000 NOAH the stage distributes to others, 3,750,000 NOAH. Contributor awards follow a guideline of 40% of each stage's distribution, which governance may exceed with an explanation. The gate and the guideline are governance policy, not genesis configuration.

B.4 Founder voting power

Governance requires a 50% quorum, 66.7% approval for ordinary proposals and 75% for expedited ones, and a 33.4% veto threshold (Table 7). These percentages retain the SDK's tally rules, including the treatment of abstentions; they do not replace stake weighting with a separate seat-counting implementation.

This ten-seat illustration assumes all relevant stake votes, no abstentions or veto votes, all non-founder stake supports the proposal, and no other stake changes. It isolates the effect of public stake; actual outcomes follow the SDK's complete tally rules and how delegated votes are exercised.

Let $F = 22{,}500{,}000$ NOAH of founding stake controlled by the founder, $O = 45{,}000{,}000$ held by the other founding seats, a the founder's additional paid awards that are bonded and voted, and p independently exercised public stake. At the ordinary threshold $\theta = 0.667$, the supporting share against the founder's opposition is:

$$\text{supporting share} = \frac{O + p}{F + O + a + p}$$

$$\text{supporting share} > \theta \quad \text{requires:}$$

$$p > \frac{\theta(F + O + a) - O}{1 - \theta}$$

With $a = 0$, the threshold is crossed at approximately 67,568 NOAH of public support, conventionally rounded to about 68,000 in the launch illustration. This small amount reflects the difference between an exact two-thirds share and the stated 66.7% approval threshold. It is not a prediction of turnout or a guarantee that public holders oppose the founder.

Each additionally bonded and voted NOAH of the founder's awards raises that illustrative public-stake requirement by approximately 2.003 NOAH. If another founding seat's 5,000,000 NOAH also opposes, the requirement rises by approximately 15,015,015 NOAH, before the initial offset. The rounded descriptions of twice the paid awards and 15 million

per additional seat are approximations under these assumptions, not separate protocol rules.

At launch the founder cannot meet the 50% quorum alone. Bonded awards lift the founder towards it: once they exceed the founder's seat plus independently exercised public stake, which with all 25,000,000 NOAH of awards paid and bonded takes public stake under 2,500,000 NOAH, the founder alone meets quorum and passes any proposal that no other stake votes against. Against the other seats voting no, the founder's 51.35% stays below the 66.7% threshold, which is the full-turnout case the illustration above assumes. The blocking position changes with grant payment, bonding, turnout, delegation, and coalitions. Unpaid escrow has no vote.

B.5 Committee appointments

All mandates are deliberately empty in genesis. The launch policy fixes the intended appointment triggers; actual addresses, members, and mandate windows are enacted through later governance proposals.

Table 10. Appointment triggers

Committee	Appointment trigger
Security	First governance cycle
Asset emergency	First governance cycle; separate account and, where possible, separate members
Economic policy	After the first expansions; exposure weights remain zero until calibration
Claims	Once Insurance has a charter specifying covered perils
Conversion	At or after the opening of conversion, when governance chooses to delegate conversion policy
Reserve	When a deployment is approved
Grants	Before the first member tranche
Registrar	By the grants committee with the first tranche; cleared by every grants committee appointment

Appointments use dedicated hardware-backed keys and one-chain-year terms, followed by reappointment. The initial economic corridor permits reward targets from half to twice the seat-scaled launch values and fund ratios from half to one and a half times their launch values, with Insurance capped at 10%. Exposure weights remain zero until calibration authorises their use.

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